Qns	Answer	Marks
2(ai)	The increase in internal energy of a system is the sum of external work done on the system and the heat supplied to the system.	**B1** **B1**
2(aii)	When roasting, heat is supplied to the potato, thus Q_{to} is positive. W_{on} is negligible since there is little change in volume of the potato) / W_{on} is positive since cooked potato becomes smaller in volume Therefore, $\Delta U_{increase}$ is positive. OR W_{on} is negative but magnitude of Q is larger than magnitude of W Therefore, $\Delta U_{increase}$ is positive. $\Delta U = \Delta KE + \Delta PE$ ΔPE is negligible as there is no change in state of the potato, therefore, ΔKE is positive. **OR** ΔPE is positive as water from the potato is changed to gaseous state but ΔKE is also positive. Since KE is directly proportional to (thermodynamic) temperature, the temperature of the potato increases.	**B1** **B1** **B1**
2(b)(i)	$\Delta U = Q + W$ Constant volume, therefore W = 0 J $\Delta U = Q$ $= -55$ J	 **B1** **A1**
2(b)(ii)	Process A to B takes place at constant temperature, therefore $U_A = U_B \Rightarrow \Delta U_{AB} = 0$ For cyclic process, $\Delta U_{ABCA} = 0$ $\Delta U_{AB} + \Delta U_{BC} + \Delta U_{CA} = 0$ $\Delta U_{CA} = 55$ Process C to A is an adiabatic process, therefore $Q_{to} = 0$ J $\Delta U_{CA} = W_{on} = 55$ $W_{by} = -55$	**B1** **B1** **A1**

Qns	Answer	Marks